

Chapter 4

Financial Reporting and Analysis



Financial Accounting, 11e

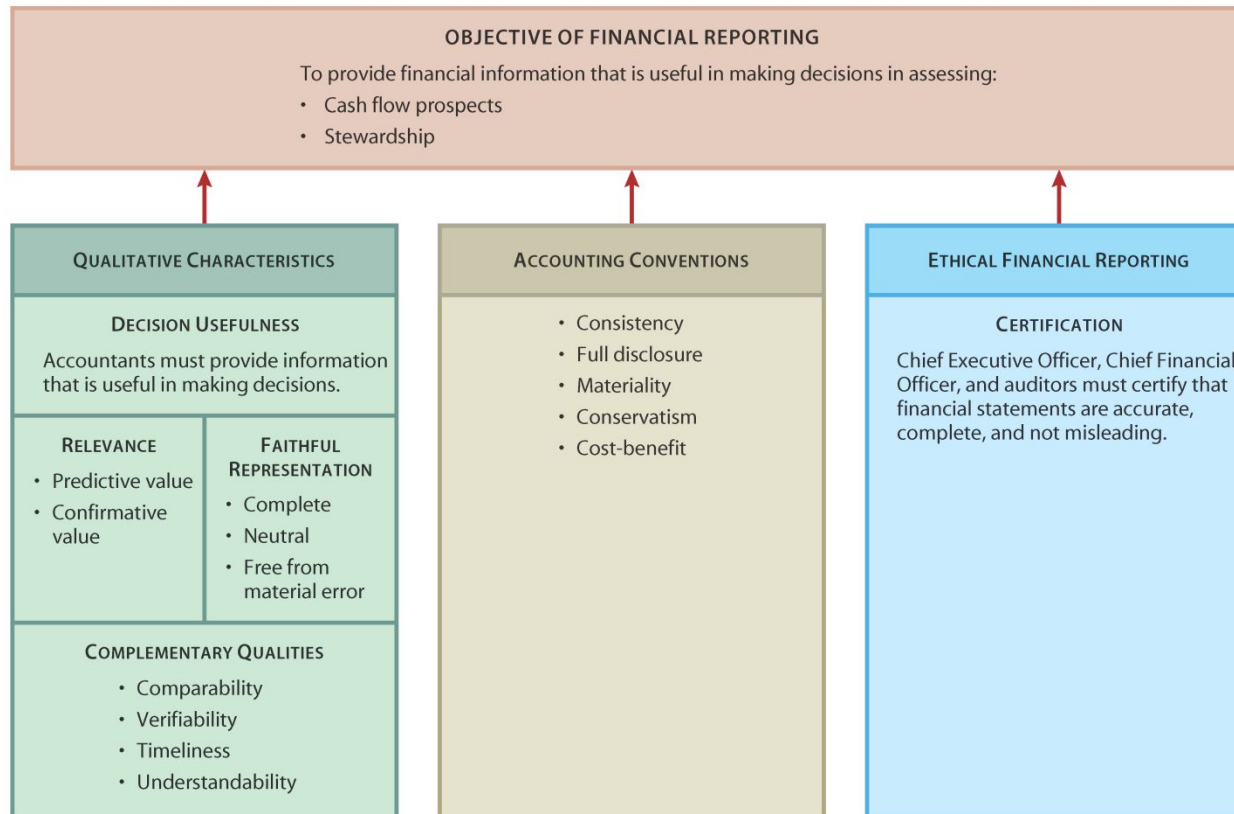
Learning Objectives

- LO 1** Describe the objective of financial reporting and identify the qualitative characteristics, conventions, and ethical considerations of accounting information.
- LO 2** Define and describe the conventions of consistency, full disclosure, materiality, conservatism, and cost-benefit.
- LO 3** Identify and describe the basic components of a classified balance sheet.
- LO 4** Describe the features of multistep and single-step classified income statements.
- LO 5** Use financial ratios to evaluate a company's performance.

Foundations of Financial Reporting

- Objective of Financial Reporting
 - Qualitative Characteristics of Accounting Information
 - ▢ **Qualitative characteristics**: Standards established by the FASB to facilitate interpretation of accounting information
 - Accounting Conventions
 - ▢ **Accounting conventions**: Constraints used in preparing financial statements
 - Ethical Financial Reporting

Factors Affecting Financial Reporting



Qualitative Characteristics

- **Relevance**: Information should have a direct bearing on a decision.
- **Faithful representation**: Financial information is complete, neutral, and free from material error.
- **Comparability**: Users can identify similarities and differences between two sets of financial data.
- **Verifiability**: Users are assured that information as presented can be substantiated.
- **Timeliness**: The quality that enables users to receive information in time to influence their decisions.
- **Understandability**: Enables users to comprehend the meaning of the information.



Stop & Apply

The numbered items below are important concepts or terms in financial accounting. Match the number these items to the letter of the category in which it belongs.

- | | |
|---|--|
| _____ 1. Furnishing information that is useful in assessing cash flow prospects | a. Decision makers (users of accounting information) |
| _____ 2. Verifiability | b. Objective of accounting information |
| _____ 3. Relevance | c. Accounting measurement considerations |
| _____ 4. Assess stewardship | d. Accounting recordkeeping considerations |
| _____ 5. Faithful representation | e. Qualitative characteristics |
| _____ 6. Recognition | |
| _____ 7. Investors | |
| _____ 8. Predictive value | |
| _____ 9. Management | |
| _____ 10. Valuation | |
| _____ 11. Internal accounting control | |
| _____ 12. Furnishing information that is useful to investors and creditors | |

SOLUTION

1. b; 2. e; 3. e; 4. b; 5. e; 6. c; 7. a;
8. e; 9. a; 10. c; 11. d; 12. b

Accounting Conventions for Preparing Financial Statements

- **Consistency**: Once a company has adopted an accounting procedure, it must use it from one period to the next unless a note to the financial statements informs users of a change in procedure.
- **Full disclosure** (or **transparency**): Requires that financial statements present all the information relevant to users' understanding of the statements.
- **Materiality**: The relative importance of an item or event; an item or event is material if there is a reasonable expectation that knowing about it would influence the decisions of users of financial statements.

Conventions

- **Conservatism:** When faced with choosing between two equally acceptable procedures or estimates, accountants should choose the one that is least likely to overstate assets and income.
- **Cost-benefit:** The benefits to be gained from providing accounting information should be greater than the costs of providing it.



Focus on International Practices



How Will Convergence of U.S. GAAP with IFRS Affect Accounting Conventions?

The FASB and the IASB are working toward merging U.S. generally accepted accounting principles (GAAP) with international financial reporting standards (IFRS). Their goal is “to increase the international comparability and the quality of standards used in the United States [which] is consistent with the FASB’s obligation to its domestic constituents, who benefit from comparability across national borders.”¹⁰ If IFRS are merged with GAAP, the comparability convention may not be the only account-

ing convention affected. For instance, conservatism, which has been the bedrock of U.S. accounting practice for many decades, might no longer be part of the conceptual framework. The practice under IFRS of writing up the value of an asset, such as inventory or equipment, that has increased in fair value and recording it as income violates the conservatism convention under GAAP. Thus, the convergence of IFRS with GAAP may well influence the way accountants in the United States analyze financial statements.



Stop & Apply

Match each item on the left below to the related accounting convention on the right.

- | | |
|--|--------------------|
| _____ 1. A note to the financial statements explains the company's method of revenue recognition. | a. Consistency |
| _____ 2. Inventory is accounted for at its market value, which is less than its original cost. | b. Full disclosure |
| _____ 3. A company uses the same method of revenue recognition year after year. | c. Materiality |
| _____ 4. Several accounts are grouped into one category because the total amount of each account is small. | d. Conservatism |
| _____ 5. A company does not keep detailed records of certain operations because the information gained from the detail is not deemed useful. | e. Cost-benefit |

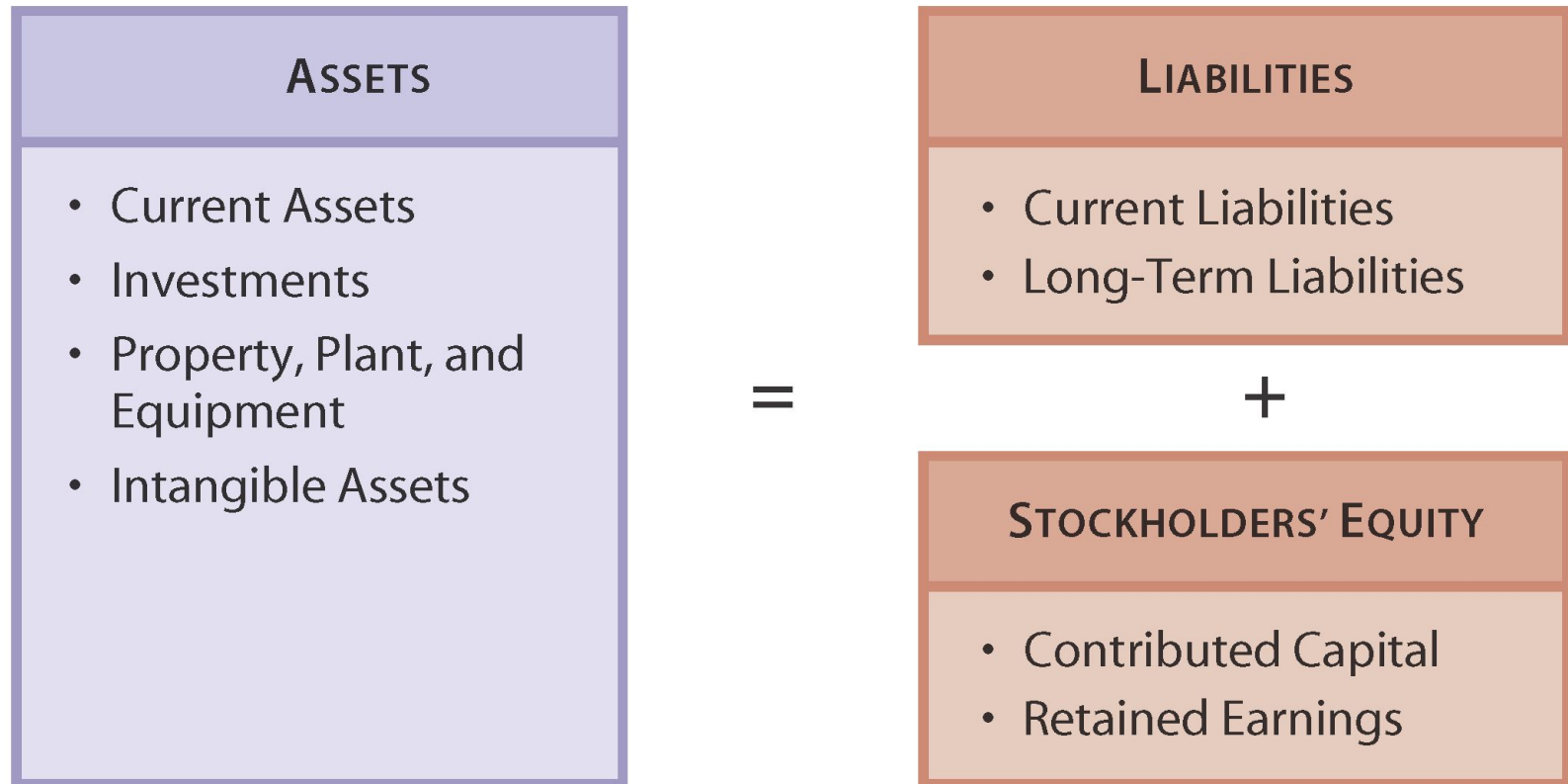
SOLUTION

1. b; 2. d; 3. a; 4. c; 5. e

Classified Balance Sheet

- General-purpose external financial statements that are divided into subcategories are called **classified financial statements**.

Classified Balance Sheet



Assets

(slide 1 of 2)

- **Current assets:** Cash and other assets that a company can reasonably expect to convert to cash, sell, or consume within one year or its *normal operating cycle*, whichever is longer.
 - **Normal operating cycle:** Average time needed to go from spending cash to receiving cash.

Assets

(slide 2 of 2)

- **Investments**: Usually long-term assets not used in normal business operations and not planned to convert to cash within the next year.
- **Property, plant, and equipment**: Tangible long-term assets used in day-to-day operations.
- **Intangible assets**: Long-term assets with no physical substance whose value stems from the rights or privileges accruing to their owners.
 - For simplicity, some companies group investments, intangible assets, and other miscellaneous assets into a single category called **other assets**.

Liabilities

- **Current liabilities:** Obligations that must be satisfied within one year or within the company's normal operating cycle, whichever is longer.
- **Long-term liabilities:** Debts that fall due more than one year in the future or beyond the normal operating cycle and are thus paid out of noncurrent assets.

Stockholders' Equity

- **Contributed capital:** The amount that stockholders invest in the business.
- **Retained earnings:** The earnings of the corporation less any losses, dividends, or transfers to contributed capital.

Classified Balance Sheet for Martin Auto Parts Corporation

Martin Auto Parts Corporation			
Balance Sheet			
December 31, 2011			
Assets			
Current assets			
Cash		\$ 41,440	
Short-term investments		28,000	
Notes receivable		32,000	
Accounts receivable		141,200	
Merchandise inventory		191,600	
Prepaid insurance		26,400	
Supplies		<u>6,784</u>	
Total current assets			\$467,424
Investments			
Land held for future use			50,000
Property, plant, and equipment			
Land		\$ 18,000	
Building	\$ 82,600		
Less accumulated depreciation	<u>34,560</u>		48,040
Equipment	<u>\$108,000</u>		
Less accumulated depreciation	<u>57,800</u>		<u>50,200</u>
Total property, plant, and equipment			116,240
Intangible assets			
Trademark			<u>2,000</u>
Total assets			<u>\$635,664</u>
Liabilities			
Current liabilities			
Notes payable		\$ 60,000	
Accounts payable		102,732	
Salaries payable		<u>8,000</u>	
Total current liabilities			\$ 170,732
Long-term liabilities			
Mortgage payable			<u>71,200</u>
Total liabilities			\$241,932
Stockholders' Equity			
Contributed capital			
Common stock, \$10 par value, 20,000 shares authorized, issued, and outstanding		\$200,000	
Additional paid-in capital		<u>40,000</u>	
Total contributed capital			\$240,000
Retained earnings			
Total stockholders' equity		<u>153,732</u>	<u>393,732</u>
Total liabilities and stockholders' equity			<u>\$635,664</u>

Owner's Equity and Partners' Equity

- The equity section of a sole proprietorship's balance sheet shows the capital in the owner's name at an amount equal to the net assets of the company.
 - No legal separation between the owner and the business, so no need to separate contributed capital from earnings retained.
- The Capital account is
 - *increased* by the owner's investments and net income.
 - *decreased* by net losses and withdrawals of assets by the owner.
- The equity section of a partnership's balance sheet is called *partners' equity*.

Classified Balance Sheet for Dell Computer Corporation

Dell Computer Corporation
Consolidated Statement of Financial Position
(In millions)

	January 29, 2010	January 30, 2009
Assets		
Current assets:		
Cash and cash equivalents	\$10,635	\$ 8,352
Short-term investments	373	740
Accounts receivable, net	5,837	4,731
Financing receivables, net	2,706	1,712
Inventories, net	1,051	867
Other current assets	3,643	3,749
Total current assets	<u>\$24,245</u>	<u>\$20,151</u>
Property, plant, and equipment, net	2,181	2,277
Investments	781	454
Long-term financing receivables, net	332	500
Goodwill	4,074	1,737
Purchased intangible assets, net	1,694	724
Other non-current assets	345	657
Total assets	<u>\$33,652</u>	<u>\$26,500</u>
Liabilities and Stockholders' Equity		
Current liabilities:		
Short-term debt	\$ 663	\$ 113
Accounts payable	11,373	8,309
Accrued and other	3,884	3,736
Short-term deferred services revenue	3,040	2,701
Total current liabilities	<u>\$18,960</u>	<u>\$14,859</u>
Long-term debt	3,417	1,898
Long-term deferred services revenue	3,029	3,000
Other non-current liabilities	2,605	2,472
Total liabilities	<u>\$28,011</u>	<u>\$22,229</u>
Stockholders' equity:		
Preferred stock and capital in excess of \$.01 par value; shares issued and outstanding:	—	—
Common stock and capital in excess of \$.01 par value; shares authorized: 7,000; shares issued: 3,351 and 3,338, respectively; shares outstanding: 1,957 and 1,944, respectively	11,472	11,189
Treasury stock, at cost; 919 shares	(27,904)	(27,904)
Retained earnings	22,110	20,677
Other comprehensive loss	(37)	309
Total stockholders' equity	<u>5,641</u>	<u>4,271</u>
Total liabilities and stockholders' equity	<u>\$33,652</u>	<u>\$26,500</u>

Source: Adapted from Dell Computer Corporation, *Annual Report*, 2010.



Focus on International Practices

IFRS

IASB Proposes Change in Format of Financial Statements

In the United States, classified financial statements have been used for more than a century, and accountants and most people engaged in business are very familiar with their format. However, the format of financial statements may change in the near future. The International Accounting Standards Board (IASB) is considering reorganizing the balance sheet and income statement so that their formats resemble the one used in the statement of cash flows. Under the proposal, both statements would include the

categories now found on the statement of cash flows: operating, investing, and financing activities. The balance sheet form that equates total assets with liabilities and stockholders' equity ($A = L + SE$) would be replaced by a form in which each category of liabilities is netted against its corresponding asset category. For example, current operating liabilities would be subtracted from current operating assets, and long-term debt would be subtracted from long-term assets on the asset side of the balance sheet.

iStock Photo



Stop & Apply

The numbered items are account titles. The lettered items below are categories that appear on a balance sheet. Match each account to the letter of the appropriate category or indicate that it does not appear on the balance sheet.

- ___ 1. Trademark
- ___ 2. Marketable Securities
- ___ 3. Land Held for Future Use
- ___ 4. Taxes Payable
- ___ 5. Bond Payable in Five Years
- ___ 6. Common Stock
- ___ 7. Land Used in Operations
- ___ 8. Accumulated Depreciation
- ___ 9. Accounts Receivable
- ___ 10. Interest Expense
- ___ 11. Unearned Revenue
- ___ 12. Prepaid Rent

- a. Current assets
- b. Investments
- c. Property, plant, and equipment
- d. Intangible assets
- e. Current liabilities
- f. Long-term liabilities
- g. Stockholders' equity
- h. Not on balance sheet

SOLUTION

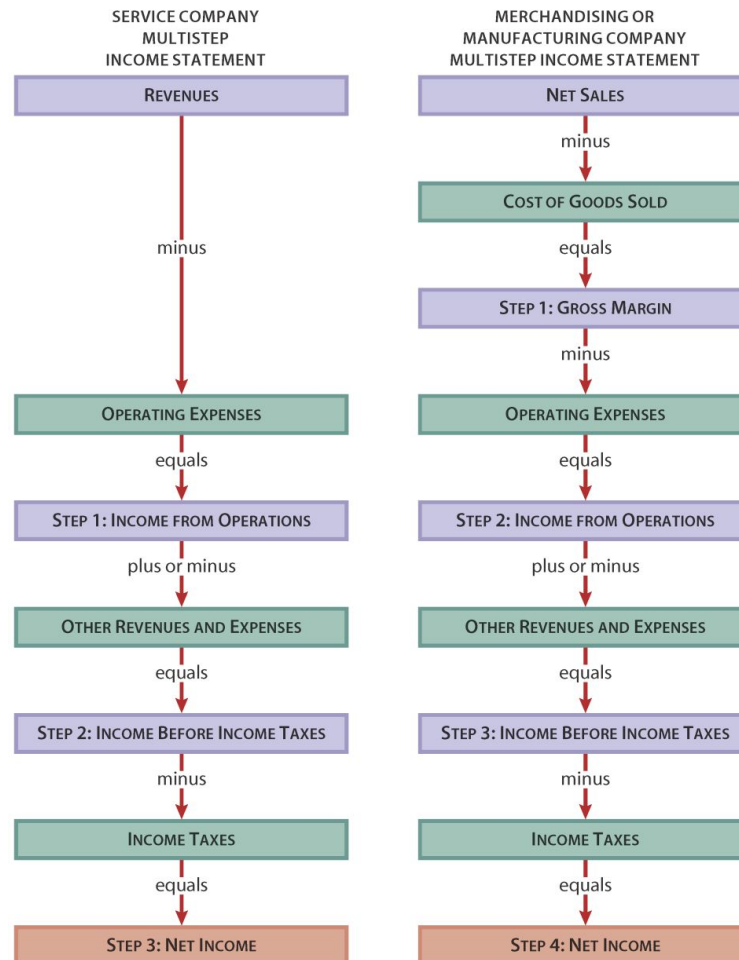
1. d; 2. a; 3. b; 4. e; 5. f; 6. g; 7. c;
8. c; 9. a; 10. h; 11. e; 12. a

Forms of the Income Statement:

Multistep Income Statement

- Forms of the income statement:
 - Multi-step income statement
 - Single-step income statement
- **Multistep Income Statement** goes through a series of steps, or subtotals, to arrive at net income.
 - Format differs depending on type of company:
 - Service company provides services as opposed to products
 - **Merchandising company** buys and sells products
 - **Manufacturing company** makes and sells products

A Comparison of the Components of Multistep Income Statements for Service and Merchandising or Manufacturing Companies



Net Sales

- **Net sales:** Gross sales less sales returns and allowances.
 - **Gross sales:** The total revenue from cash and credit sales recorded during an accounting period.
 - **Sales returns and allowances:** Cash refunds and credits on account.

Cost of Goods Sold & Gross Margin

- **Cost of goods sold:** The amount a merchandiser paid for the merchandise it sold during an accounting period.
- **Gross margin:** The difference between net sales and the cost of goods sold.

Operating Expenses

- **Operating expenses:** Expenses incurred in running a business other than the cost of goods sold.
 - **Selling expenses:** The costs of storing and preparing goods for sale; preparing displays, advertising, and otherwise promoting sales; and delivering goods if seller has agreed to pay the cost of delivery.
 - **General and administrative expenses:** Expenses for accounting, personnel, credit checking, collections, and any other expenses that apply to overall operations.
- **Income from operations:** The income from a company's main business.
- **Other revenues and expenses:** Expenses *not* related to a company's operating activities.

Income Before Taxes & Income Taxes

- **Income before income taxes:** Amount a company has earned from all activities—operating and nonoperating—before taking into account the amount of income taxes it incurred.
- **Income taxes:** The expense for federal, state, and local taxes on income.

Net Income & Earnings per Share

- **Net income**: The amount of earnings that accrue to stockholders.
 - **Earnings per share (net income per share)**: The net income earned on each share of common stock.

$$\text{Earnings per share} = \text{Net Income Average} \div \text{Shares of Common Stock Outstanding}$$

Multistep Income Statement for Martin Auto Parts Corporation

Martin Auto Parts Corporation Income Statement For the Year Ended December 31, 2011			
Step 1	Net sales		\$1,248,624
	Cost of goods sold		<u>815,040</u>
	Gross margin		\$ 433,584
Step 2	Operating expenses		
	Selling expenses	\$219,120	
	General and administrative expenses	<u>138,016</u>	
	Total operating expenses		<u>357,136</u>
	Income from operations		\$ 76,448
Step 3	Other revenues and expenses		
	Interest income	\$ 5,600	
	Less interest expense	<u>10,524</u>	
	Excess of other expenses over other revenues		<u>4,924</u>
	Income before income taxes		\$ 71,524
Step 4	Income taxes		<u>13,524</u>
	Net income		\$ 58,000
	Earnings per share		<u><u>\$ 2.90</u></u>

Multistep Income Statement for Dell Computer Corporation

Dell Computer Corporation Consolidated Statement of Income (In millions, except per share amounts)			
	Fiscal Year Ended		
	January 29, 2010	January 30, 2009	February 1, 2008
Net revenue	\$52,902	\$61,101	\$61,133
Cost of revenue	43,641	50,144	49,462
Gross margin	<u>9,261</u>	<u>10,957</u>	<u>11,671</u>
Operating expenses:			
Selling, general, and administrative	6,465	7,102	7,538
In-process research and development	—	2	83
Research, development, and engineering	624	663	610
Total operating expenses	<u>7,089</u>	<u>7,767</u>	<u>8,231</u>
Operating income	2,172	3,190	3,440
Interest and other, net	<u>(148)</u>	<u>134</u>	<u>387</u>
Income before income taxes	2,024	3,324	3,827
Income tax provision	591	846	880
Net income	<u>\$ 1,433</u>	<u>\$ 2,478</u>	<u>\$ 2,947</u>
Earnings per share*	<u>\$ 0.73</u>	<u>\$ 1.25</u>	<u>\$ 1.33</u>

*Basic

Source: Dell Computer Corporation, *Annual Report*, 2010.

Forms of the Income Statement:

Single-Step Income Statement

- **Single-step income statement:** Income before income taxes is derived in a single step by putting the
 - Major categories of *revenues* in the first part of the statement.
 - Major categories of *costs and expenses* in the second part of the statement.

Single-Step Income Statement for Martin Auto Parts Corporation

Martin Auto Parts Corporation
Income Statement
For the Year Ended December 31, 2011

Revenues

Net sales		\$ 1,248,624
Interest income		5,600
Total revenues		<u>\$ 1,254,224</u>

Costs and expenses

Cost of goods sold	\$815,040	
Selling expenses	219,120	
General and administrative expenses	138,016	
Interest expense	<u>10,524</u>	
Total costs and expenses		<u>1,182,700</u>

Income before income taxes \$ 71,524

Income taxes 13,524

Net income \$ 58,000

Earnings per share \$ 2.90



Stop & Apply

The numbered items are account titles. The lettered items below are categories that appear on a multistep income statement. Match each account to the letter of the appropriate category or indicate that it is not on the income statement.

- | | |
|---------------------------------------|--|
| _____ 1. Sales Returns and Allowances | a. Net sales |
| _____ 2. Cost of Goods Sold | b. Cost of goods sold |
| _____ 3. Dividend Income | c. Selling expenses |
| _____ 4. Delivery Expense | d. General and administrative expenses |
| _____ 5. Office Salaries Expense | e. Other revenues and expenses |
| _____ 6. Wages Payable | f. Not on income statement |
| _____ 7. Sales Salaries Expense | |
| _____ 8. Advertising Expense | |
| _____ 9. Interest Expense | |
| _____ 10. Commissions Expense | |

SOLUTION

1. a; 2. b; 3. e; 4. c; 5. d; 6. f; 7. c;
8. c; 9. e; 10. c

Using Financial Ratios for Performance Evaluation

- Critical Financial Ratios and Financial Statement Elements:
 - **Profit margin**: Percentage of each sales dollar that results in net income.
 - **Asset turnover**: How efficiently assets are used to produce sales
 - **Cash flow yield**: How much cash operations generate in relation to net income.
 - **Debt to equity**: A company's strategy for financing its operations.

Profit Margin for Martin Auto Parts

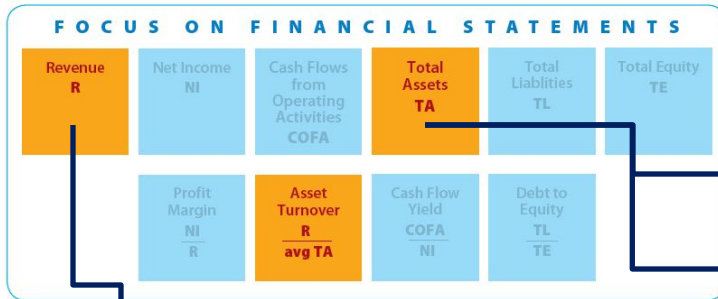


Martin Auto Parts Corporation
Income Statement
For the Year Ended December 31, 2011

Step 1	Net sales		\$1,248,624
	Cost of goods sold		815,040
	Gross margin		\$ 433,584
Step 2	Operating expenses		
	Selling expenses	\$219,120	
	General and administrative expenses	138,016	
	Total operating expenses		357,136
	Income from operations		\$ 76,448
Step 3	Other revenues and expenses		
	Interest income	\$ 5,600	
	Less interest expense	10,524	
	Excess of other expenses over other revenues		4,924
	Income before income taxes		\$ 71,524
Step 4	Income taxes		13,524
	Net income		\$ 58,000
	Earnings per share		\$ 2.90

$$\begin{aligned}
 \text{Profit Margin} &= \frac{\text{Net Income}}{\text{Revenue}} \\
 &= \frac{\$58,000}{\$1,248,624} \\
 &= 0.046, \text{ or } 4.6\%
 \end{aligned}$$

Asset Turnover for Martin Auto Parts



Martin Auto Parts Corporation
Balance Sheet
December 31, 2011

Total assets	<u>\$635,664</u>
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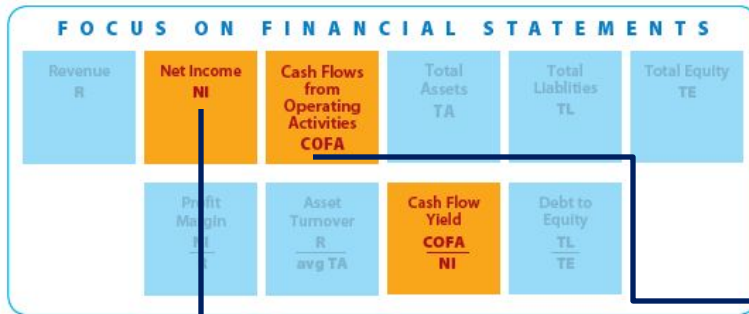
If Martin had assets of \$594,480 at the beginning of the year:

Martin Auto Parts Corporation
Income Statement
For the Year Ended December 31, 2011

Step 1	Net sales	\$1,248,624
	Cost of goods sold	815,040
	Gross margin	\$ 433,584
Step 2	Operating expenses	
	Selling expenses	\$219,120
	General and administrative expenses	138,016
	Total operating expenses	<u>357,136</u>
	Income from operations	\$ 76,448
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	Interest income	\$ 5,600
	Less interest expense	<u>10,524</u>
	Excess of other expenses over other revenues	<u>4,924</u>
	Income before income taxes	\$ 71,524
Step 4	Income taxes	13,524
	Net income	<u>\$ 58,000</u>
	Earnings per share	<u>\$ 2.90</u>

$$\begin{aligned}
 \text{Asset Turnover} &= \frac{\text{Revenue}}{\text{Average Total Assets}} \\
 &= \frac{\$1,248,624}{(\$635,664 + \$594,480) \div 2} \\
 &= \frac{\$1,248,624}{\$615,072} \\
 &= 2.0 \text{ times}
 \end{aligned}$$

Cash Flow Yield for Martin Auto Parts



If Martin's cash flow from operating activities is \$72,500:

Martin Auto Parts Corporation
Income Statement
For the Year Ended December 31, 2011

Step 1	Net sales		\$1,248,624
	Cost of goods sold		815,040
	Gross margin		\$ 433,584
Step 2	Operating expenses		
	Selling expenses	\$219,120	
	General and administrative expenses	138,016	
	Total operating expenses		357,136
	Income from operations		\$ 76,448
Step 3	Other revenues and expenses		
	Interest income	\$ 5,600	
	Less interest expense	10,524	
	Excess of other expenses over other revenues		4,924
	Income before income taxes		\$ 71,524
	Income taxes		13,524
Step 4	Net income		\$ 58,000
	Earnings per share		\$ 2.90

$$\text{Cash Flow Yield} = \frac{\text{Cash Flows from Operating Activities}}{\text{Net Income}}$$

$$= \frac{\$72,500}{\$58,000}$$

$$= 1.3 \text{ times}$$

Debt to Equity Ratio for Martin Auto Parts

Martin Auto Parts Corporation Balance Sheet December 31, 2011		
Liabilities		
Current liabilities		
Notes payable	\$ 60,000	
Accounts payable	102,732	
Salaries payable	8,000	
Total current liabilities		\$ 170,732
Long-term liabilities		
Mortgage payable		71,200
Total liabilities		\$241,932
Stockholders' Equity		
Contributed capital		
Common stock, \$10 par value, 20,000 shares authorized, issued, and outstanding	\$200,000	
Additional paid-in capital	40,000	
Total contributed capital	\$240,000	
Retained earnings		
Total stockholders' equity	153,732	
Total liabilities and stockholders' equity		\$635,664

FOCUS ON FINANCIAL STATEMENTS					
Revenue R	Net Income NI	Cash Flows from Operating Activities COFA	Total Assets TA	Total Liabilities TL	Total Equity TE
	Profit Margin $\frac{NI}{R}$	Asset Turnover $\frac{R}{\text{avg TA}}$	Cash Flow Yield $\frac{COFA}{NI}$	Debt to Equity $\frac{TL}{TE}$	

$$\begin{aligned} \text{Debt to Equity} &= \frac{\text{Total Liabilities}}{\text{Total Equity}} \\ &= \frac{\$241,932}{\$393,732} \\ &= 0.614, \text{ or } 61.4\% \end{aligned}$$

Beyond the Basics:

Additional Financial Ratios

- **Return on assets:** Relates net income from the income statement to average total assets from the balance sheet.
- **Cash return on assets:** Relates cash flows from operating activities from the statement of cash flows to average total assets from the balance sheet.
- **Return on equity:** The ratio of net income to average total equity.

Return on Assets for Martin Auto Parts

FOCUS ON FINANCIAL STATEMENTS



Martin Auto Parts Corporation

Balance Sheet

December 31, 2011

Total assets

\$635,664

Martin Auto Parts Corporation Income Statement For the Year Ended December 31, 2011

	Net sales	\$1,248,624
Step 1	Cost of goods sold	<u>815,040</u>
	Gross margin	\$ 433,584
	Operating expenses	
Step 2	Selling expenses	\$219,120
	General and administrative expenses	<u>138,016</u>
	Total operating expenses	<u>357,136</u>
	Income from operations	\$ 76,448
	Other revenues and expenses	
Step 3	Interest income	\$ 5,600
	Less interest expense	<u>10,524</u>
	Excess of other expenses over other revenues	<u>4,924</u>
	Income before income taxes	\$ 71,524
	Income taxes	<u>13,524</u>
Step 4	Net income	<u>\$ 58,000</u>
	Earnings per share	<u>\$ 2.90</u>

If Martin had assets of \$594,480 at the beginning of the year:

$$\begin{aligned}
 \text{Return on Assets} &= \frac{\text{Net Income}}{\text{Average Total Assets}} \\
 &= \frac{\$58,000}{(\$635,664 + \$594,480) \div 2} \\
 &= \frac{\$58,000}{\$615,072} \\
 &= 0.094, \text{ or } 9.4\%
 \end{aligned}$$

Cash Return on Assets for Martin Auto Parts

FOCUS ON FINANCIAL STATEMENTS

Revenue R	Net Income NI	Cash Flows from Operating Activities COFA	Total Assets TA	Total Liabilities TL	Total Equity TE
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If Martin had cash flows from operating activities of \$72,500 and had assets of \$594,480 at the beginning of the year:

$$\begin{aligned}
 \text{Cash Return on Assets} &= \frac{\text{Cash Flows from Operating Activities}}{\text{Average Total Assets}} \\
 &= \frac{\$72,500}{(\$635,664 + \$594,480) \div 2} \\
 &= \frac{\$72,500}{\$615,072} \\
 &= 0.118, \text{ or } 11.8\%
 \end{aligned}$$

Martin Auto Parts Corporation Balance Sheet December 31, 2011

Total assets

\$635,664

Return on Equity for Martin Auto Parts

FOCUS ON FINANCIAL STATEMENTS

Revenue R	Net Income NI	Cash Flows from Operating Activities COFA	Total Assets TA	Total Liabilities TL	Total Equity TE
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Martin Auto Parts Corporation

Balance Sheet

December 31, 2011

Total stockholders' equity

393,732

Martin Auto Parts Corporation

Income Statement

For the Year Ended December 31, 2011

Step 1	Net sales		\$1,248,624
	Cost of goods sold		815,040
	Gross margin		\$ 433,584
Step 2	Operating expenses		
	Selling expenses	\$219,120	
	General and administrative expenses	138,016	
	Total operating expenses	357,136	
	Income from operations		\$ 76,448
Step 3	Other revenues and expenses		
	Interest income	\$ 5,600	
	Less interest expense	10,524	
	Excess of other expenses over other revenues	4,924	
	Income before income taxes		\$ 71,524
Step 4	Income taxes		13,524
	Net income		\$ 58,000
	Earnings per share		\$ 2.90

If Martin had \$402,212 in equity at the beginning of the year:

$$\begin{aligned}
 \text{Return on Equity} &= \frac{\text{Net Income}}{\text{Average Total Equity}} \\
 &= \frac{\$58,000}{(\$393,732 + \$402,212) \div 2} \\
 &= \frac{\$58,000}{\$397,972} \\
 &= 0.146, \text{ or } 14.6\%
 \end{aligned}$$



Stop & Apply

The following end-of-year amounts are from the financial statements of Roth Company: total assets, \$220,000; total liabilities, \$80,000; stockholders' equity, \$140,000; net sales, \$300,000; net income, \$18,000; and cash flows from operating activities, \$24,000. One year ago, total assets were \$180,000, and stockholders' equity was \$122,000. Compute Roth's (1) profit margin, (2) asset turnover, (3) cash flow yield, (4) debt to equity ratio, (5) return on assets, (6) cash return on assets, and (7) return on equity. (Note: Round to one decimal place.)

SOLUTION

- (1) Profit margin: $\$18,000 \div \$300,000 = 0.06$, or 6.0%
- (2) Asset turnover: $\$300,000 \div [(\$180,000 + \$220,000) \div 2] = 1.5$ times
- (3) Cash flow yield: $\$24,000 \div \$18,000 = 1.3$ times
- (4) Debt to equity ratio: $\$80,000 \div \$140,000 = 0.571$, or 57.1%
- (5) Return on assets: $\$18,000 \div [(\$180,000 + \$220,000) \div 2] = 0.09$, or 9.0%
- (6) Cash return on assets: $\$24,000 \div [(\$180,000 + \$220,000) \div 2] = 0.12$, or 12.0%
- (7) Return on equity: $\$18,000 \div [(\$122,000 + \$140,000) \div 2] = 0.137$, or 13.7%